FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Mathematics — Class IX

Syllabus: Unit 1 — Real Numbers | Unit 2 — Logarithms | Unit 4 — Factorization & Algebraic Manipulation

Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry

ANSWER KEY (Version 2)

Total Marks: 65 **Time Allowed: 2 Hours 30 Minutes** **Version: 2**

SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

#	Answer	Explanation
1	**B**	The **Quotient Rule of exponents** states $a^m \div a^n = a^{m-n}$. Product rule handles multiplication; Power rule raises a power to a power; Negative exponent gives $1/a^n$.
2	**C**	For $a > 0$, $\sqrt{a}$ is always a **non-negative real number**. It cannot be negative (distance concept). It may or may not be irrational, and is not necessarily an integer.
3	**B**	$0.000307 = 3.07 \times 10^{-4}$. The decimal moves 4 places right to place it after the first nonzero digit, so the exponent is **−4**. Option D is not standard form (d must satisfy $1 \le d < 10$).
4	**C**	$\log_3 81 = y$ means $3^y = 81 = 3^4$, so $\mathbf{y = 4}$. Distractor 3 comes from thinking 3×3=9 not 81; distractor 9 from wrong base arithmetic.
5	**B**	$18 = 2 \times 3^2$, so $\log 18 = \log 2 + \log 3^2 = \mathbf{\log 2 + 2 \log 3}$ (product law + power law). Common error: writing $\log(2\times3) = \log 2 + \log 3$ only, missing the square.
6	**A**	$x^2 - 9x + 20$: need two numbers that multiply to 20 and add to −9 ⇒ **−4 and −5**. So factors are $(x-4)(x-5)$. Check: $(x-4)(x-5) = x^2-9x+20$. √
7	**C**	The fundamental theorem states **HCF × LCM = P × Q** for any two polynomials P and Q. This is used to find the second polynomial when one, HCF, and LCM are known.
8	**B**	Standard form of a linear equation in one variable is $\mathbf{ax + b = 0}$ where $a, b \in R$ and $a \neq 0$. $ax^2+b=0$ is quadratic; $ax+by=0$ has two variables.
9	**A**	$\lvert 3x-6\rvert = 9$ gives two equations: $3x-6 = 9 \Rightarrow x=5$, or $3x-6 = -9 \Rightarrow 3x=-3 \Rightarrow x=-1$. Solution set = **{5, −1}**.
10	**C**	$d = \sqrt{(5^2 + 12^2)} = \sqrt{(25+144)} = \sqrt{169} = \mathbf{13}$. This is the classic 5-12-13 Pythagorean triple. Distractor 17 = 5+12 (wrong: not sum of sides).
11	**B**	Midpoint = $((4+(-2))/2, (-2+8)/2) = (2/2, 6/2) = \mathbf{(1, 3)}$. Common error: adding instead of averaging, giving (2, 6).
12	**B**	$x^2-4 = (x+2)(x-2)$; $x^2-x-2 = (x-2)(x+1)$. The common factor is $\mathbf{(x-2)}$. Distractor A, $(x+2)$, appears in x^2-4 but NOT in x^2-x-2 (which factors as $(x-2)(x+1)$).

SECTION B — Short Question Answers (11 × 3 = 33 Marks)

2(i) — Unit 1: Laws of Exponents **3 marks**

Simplify $(-3)^2 \cdot (-3)^{-4} \cdot (-3)^3 \div (-3)^{-2}$

Using **product rule** (multiply: add exponents): $(-3)^{2+(-4)+3} = (-3)^1$ **1 mk**

Division: $(-3)^1 \div (-3)^{-2} = (-3)^{1-(-2)} = (-3)^3$ **1 mk**

$(-3)^3 = \mathbf{-27}$ **1 mk**

OR — Radical expressions:

(i) $(-8)^{1/3} = \sqrt[3]{(-8)} = \mathbf{-2}$ (cube root of -8) **1 mk**

(ii) $(\sqrt{25})^4 = (5)^4 = \mathbf{625}$ **1 mk**

(iii) $(x^{-3})^2 \cdot (x^0)^5 = x^{-6} \cdot 1 = \mathbf{x^{-6}}$ (or $1/x^6$) **1 mk**

2(ii) — Unit 1: Real Numbers Classification **3 marks**

(a) Every integer is a rational number — **TRUE**. Any integer $n = n/1$, which is rational. **1 mk**

(b) Every irrational number is a real number — **TRUE**. Irrationals are a subset of real numbers. **1 mk**

(c) Every rational number is an integer — **FALSE**. Counter-example: 1/2 is rational but not an integer. **1 mk**

OR — Number Line:

(i) $-3½ = -3.5$: mark between -4 and -3, closer to -4. **1 mk**

(ii) $19/4 = 4.75$: mark between 4 and 5, three-quarters of the way. **1 mk**

(iii) $x \geq -2$: closed circle at -2, arrow pointing right. **1 mk**

2(iii) — Unit 2: Scientific Notation Computation **3 marks**

Numerator: $4.2 \times 10^5 \times 3.0 \times 10^{-3} = (4.2 \times 3.0) \times 10^{5+(-3)} = 12.6 \times 10^2$ **1 mk**

Divide: $(12.6 \times 10^2) \div (6.0 \times 10^4) = (12.6/6.0) \times 10^{2-4} = 2.1 \times 10^{-2}$ **1 mk**

Answer in standard scientific notation: $\mathbf{2.1 \times 10^{-2}}$ **1 mk**

OR — Notation Conversion:

(i) $7.05 \times 10^{-4} = \mathbf{0.000705}$ (move decimal 4 places left) **1 mk**

(ii) $2.31 \times 10^6 = \mathbf{2,310,000}$ (move decimal 6 places right) **1 mk**

(iii) $0.0000000512 = \mathbf{5.12 \times 10^{-8}}$ (first nonzero digit after decimal: count 8 places) **1 mk**

2(iv) — Unit 2: Logarithm — Find Unknown **3 marks**

(i) $\log_x 125 = 3 \Rightarrow x^3 = 125 = 5^3 \Rightarrow \mathbf{x = 5}$ **1 mk**

(ii) $\log_4 x = -3/2 \Rightarrow x = 4^{-3/2} = (2^2)^{-3/2} = 2^{-3} = \mathbf{1/8}$ **1 mk**

(iii) $\log_2(x^2-3) = \log_2 13 \Rightarrow x^2-3 = 13 \Rightarrow x^2 = 16 \Rightarrow \mathbf{x = \pm 4}$ (both satisfy domain: $x^2-3=13>0$) **1 mk**

OR — Expand and Combine:

Expand $\log\sqrt{(53.3/46.4)} = (1/2)[\log 53.3 - \log 46.4] = (1/2)[1.7267 - 1.6665] = (1/2)(0.0602) = \mathbf{0.0301}$ **1.5 mk**

Combine $3 \log p - 2 \log q + (1/2) \log r = \log p^3 - \log q^2 + \log r^{1/2} = \mathbf{\log(p^3\sqrt{r} / q^2)}$ **1.5 mk**

2(v) — Unit 2: Log Tables — Characteristic & Mantissa **3 marks**

(i) $4580 = 4.580 \times 10^3$; characteristic = 3; mantissa from table for 4580 = .6609; **log 4580 = 3.6609** **1 mk**

(ii) $0.00093 = 9.3 \times 10^{-4}$; characteristic = $\bar{4}$; mantissa for 93 = .9685; **log 0.00093 =** $\mathbf{\bar{4}.9685}$ **1 mk**

(iii) $54 = 5.4 \times 10^1$; characteristic = 1; mantissa for 54 = .7324; **log 54 = 1.7324** **1 mk**

OR — Antilog:

(i) antilog 2.4324: char=2, mantissa=.4324; table gives 2707; decimal point 2 places right ⇒ **270.7** **1 mk**

(ii) antilog $\bar{3}.5219$: char=−3, mantissa=.5219; table gives **3326**; 3 places left ⇒ **0.003326** **1 mk**

(iii) antilog 0.0000: char=0, mantissa=.0000; table gives 1000; **antilog 0.0000 = 1.000** **1 mk**

2(vi) — Unit 4: Factorization 3 marks

(i) $x^4 + 4m^4 = (x^2)^2 + (2m^2)^2$; add/subtract $4x^2m^2$: $= (x^2 + 2m^2)^2 - (2xm)^2 = \mathbf{(x^2 + 2m^2 + 2xm)(x^2 + 2m^2 - 2xm)}$ **1.5 mk**

(ii) $3x^2+11x+6$: product of extremes = 18; factors of 18 with sum 11: 9 and 2. $= 3x^2+9x+2x+6 = 3x(x+3)+2(x+3) = \mathbf{(x+3)(3x+2)}$ **1.5 mk**

OR — Factorize (sum/difference of cubes):

(i) $8a^3-125b^3 = (2a)^3-(5b)^3 = \mathbf{(2a-5b)(4a^2+10ab+25b^2)}$ **1.5 mk**

(ii) $27p^3-108p^2q+144pq^2-64q^3$: identify $a = 3p$, $b = 4q$.

$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

• $a^3 = 27p^3$ √ • $-3a^2b = -108p^2q$ √ • $+3ab^2 = +144pq^2$ √ • $-b^3 = -64q^3$ √

Therefore: $27p^3-108p^2q+144pq^2-64q^3 = \mathbf{(3p-4q)^3}$ **1.5 mk**

2(vii) — Unit 4: HCF by Factorization 3 marks

Factorize each: $x^2-49 = (x+7)(x-7)$ **1 mk**

$x^2-4x-21$: need factors of -21 summing to $-4 \Rightarrow -7$ and $+3$; $= (x-7)(x+3)$ **1 mk**

HCF = (x−7) (common factor with least power) **1 mk**

OR — LCM of three polynomials:

$x^2+x-6 = (x+3)(x-2)$; $x^2+x-2 = (x+2)(x-1)$; $x^2-4x+3 = (x-3)(x-1)$ **1.5 mk**

LCM = (x+3)(x−2)(x+2)(x−1)(x−3) — product of all distinct factors at highest power **1.5 mk**

2(viii) — Unit 5: Solve Linear Equation with Fractions 3 marks

$(2x+3)/5 = (3-4x)/8$; cross-multiply: $8(2x+3) = 5(3-4x)$ **1 mk**

$16x + 24 = 15 - 20x \Rightarrow 36x = -9$ **1 mk**

$\mathbf{x = -1/4}$ (check: LHS $= (2(-1/4)+3)/5 = (2.5)/5 = 0.5$; RHS $= (3+1)/8 = 0.5$ √) **1 mk**

OR — Radical Equation:

$\sqrt{(5x-4)} = \sqrt{(7x+2)}$; square both sides: $5x-4 = 7x+2$ **1 mk**

$-2x = 6 \Rightarrow x = -3$ **1 mk**

Check: $\sqrt{(5(-3)-4)} = \sqrt{(-19)}$ — not real. **Solution set = φ** (extraneous root) **1 mk**

2(ix) — Unit 5: Absolute Value Equation 3 marks

$|4x-3| = 13$ gives two cases: $4x-3 = 13$ or $4x-3 = -13$ **1 mk**

Case 1: $4x = 16 \Rightarrow x = 4$ Case 2: $4x = -10 \Rightarrow x = -5/2$ **1 mk**

Solution set = {4, −5/2} **1 mk**

OR — Compound Inequality:

Inequality 1: $3x+2 \le 11 \Rightarrow 3x \le 9 \Rightarrow x \le 3$ **1 mk**

Inequality 2: $2x-5 > -3 \Rightarrow 2x > 2 \Rightarrow x > 1$ **1 mk**

Combined (AND = intersection): $1 < x \le 3$; **Solution set =** $\mathbf{\{x \mid x \in \mathbb{R} \wedge 1 < x \le 3\}}$ **1 mk**

2(x) — Unit 7: Collinear Points Check 3 marks

$AB = \sqrt{((7-1)^2+(8-(-2))^2)} = \sqrt{(36+100)} = \sqrt{136} = 2\sqrt{34}$ **1 mk**

$BC = \sqrt{((4-7)^2+(3-8)^2)} = \sqrt{(9+25)} = \sqrt{34}$ $AC = \sqrt{((4-1)^2+(3-(-2))^2)} = \sqrt{(9+25)} = \sqrt{34}$ **1 mk**

Check: $AB = BC + AC \Rightarrow 2\sqrt{34} = \sqrt{34} + \sqrt{34} = 2\sqrt{34}$ √ **Points A, B, C are collinear.** **1 mk**

OR — Prove Square:

$AB = \sqrt{(25+4)} = \sqrt{29}$; $BC = \sqrt{(4+25)} = \sqrt{29}$; $CD = \sqrt{(25+4)} = \sqrt{29}$; $DA = \sqrt{(4+25)} = \sqrt{29}$ — all sides equal **1.5 mk**

Diagonal $AC = \sqrt{(49+9)} = \sqrt{58} = AB\sqrt{2}$ √ (diagonals equal). Also $AB^2+BC^2 = 29+29 = 58 = AC^2$ √ right angles. **ABCD is a square.** **1.5 mk**

2(xi) — Unit 7: Midpoint Application 3 marks

Midpoint P of A(−1,7) and B(7,−3): x-coord = $(-1+7)/2 = 3$ √; y-coord = $(7+(-3))/2 = 2$. So **k = 2**. 1 mk

P = (3, 2); distance PA = $\sqrt{(3-(-1))^2+(2-7)^2} = \sqrt{16+25} = \sqrt{41}$ 1 mk

PA = √41 units 1 mk

OR — Equal Distance:

d(2,3) to (3,t) = d(4,5) to (3,t): $\sqrt{1+(t-3)^2} = \sqrt{1+(t-5)^2}$ 1 mk

Square: $1+(t-3)^2 = 1+(t-5)^2 \Rightarrow (t-3)^2 = (t-5)^2$ 1 mk

$t^2-6t+9 = t^2-10t+25 \Rightarrow 4t = 16 \Rightarrow$ **t = 4** 1 mk

SECTION C — Long Question Answers (4 × 5 = 20 Marks)

3(i) — Unit 2: Logarithm Evaluation & Digits 5 marks

Given: $\log_b 2 = 0.3010$, $\log_b 3 = 0.4771$, $\log_b 5 = 0.6990$

(a)(i) $\log_b 30 = \log_b(2\times3\times5) = 0.3010+0.4771+0.6990 =$ **1.4771** 1 mk

(a)(ii) $\log_b(100/9) = \log_b 100 - \log_b 9 = 2\log_b 10 - 2\log_b 3$

Note: $\log_b 10 = \log_b(2\times5) = 0.3010+0.6990 = 1.0000$

$= 2(1.0000) - 2(0.4771) = 2.0000 - 0.9542 =$ **1.0458** 1.5 mk

(a)(iii) $\log_b(\sqrt{\sqrt[3]{450}}) = (1/6)\log_b 450 = (1/6)\log_b(2\times3^2\times5^2)$

$= (1/6)[0.3010+2(0.4771)+2(0.6990)] = (1/6)(2.6532) =$ **0.4422** 1 mk

(b) Digits in 3^{30}: $\log 3^{30} = 30 \times 0.4771 = 14.313$; characteristic = 14; digits = 14+1 = **15 digits** 1.5 mk

OR — Evaluate using log tables:

Let $x = \sqrt[3]{8.59} \times (55.6)^2 \div (2.51 \times \sqrt{2.12})$

$\log x = (1/3)\log 8.59 + 2\log 55.6 - \log 2.51 - (1/2)\log 2.12$ 1 mk

$= (1/3)(0.9340) + 2(1.7451) - (0.3997) - (1/2)(0.3263)$ 1 mk

$= 0.3113 + 3.4902 - 0.3997 - 0.1632 =$ **3.2386** 1.5 mk

antilog 3.2386 ≈ **1732** (from antilog table for .2386 ⇒ 1732) 1.5 mk

3(ii) — Unit 4: Algebraic Fractions 5 marks

(a) $[(x^2-1)/(x^2+4x+4)] \times [(x+2)/(x^2+2x-3)]$

Factorize: $x^2-1=(x+1)(x-1)$; $x^2+4x+4=(x+2)^2$; $x^2+2x-3=(x+3)(x-1)$ 1.5 mk

Cancel $(x-1)$ and $(x+2)$: $= (x+1) / [(x+2)(x+3)] =$ $\mathbf{(x+1)/(x^2+5x+6)}$ 1 mk

(b) $1/[x(x-y)] + 1/[y(x+y)]$; LCM = $xy(x-y)(x+y)$ 1 mk

$= [y(x+y) + x(x-y)] / [xy(x-y)(x+y)] = (x^2+y^2) / [xy(x^2-y^2)]$ 1.5 mk

OR — Square Root & Factorize:

(a) $4a^2+12ab+9b^2 = (2a)^2 + 2(2a)(3b) + (3b)^2 = (2a+3b)^2$; √ **= ±(2a+3b)** 1.5 mk

(b)(i) Let $u = x^2+3x$; then $(u-4)(u+5)+8 = u^2+u-12 = (u+4)(u-3) =$ $\mathbf{(x^2+3x+4)(x^2+3x-3)}$ 2 mk

(b)(ii) $8a^3-125b^3-2a+5b = (2a-5b)(4a^2+10ab+25b^2)-(2a-5b) =$ $\mathbf{(2a-5b)(4a^2+10ab+25b^2-1)}$ 1.5 mk

3(iii) — Unit 5: Linear Equation & Inequality **5 marks**

(a) $(y+1)/3 + (y+1)/2 = 2 - (y+3)/2$; multiply by LCM = 6: $2(y+1) + 3(y+1) = 12 - 3(y+3)$ **1 mk**

$5y+5 = 3-3y \Rightarrow 8y = -2$ **1 mk**

$\mathbf{y = -1/4}$ (Check: LHS = $1/4+3/8=5/8$; RHS = $2-11/8 = 5/8$ √) **1 mk**

(b) $(3/4)x - 1 \geq x + 1$; multiply by 4: $3x-4 \geq 4x+4 \Rightarrow -x \geq 8 \Rightarrow x \leq -8$ **1 mk**

Solution set = $\{x \mid x \in R \wedge x \leq -8\}$ **1 mk**

OR — Absolute Value & Compound Inequality:

(a) $|10-x|/5 = |2x-5|/2 \Rightarrow 2|10-x| = 5|2x-5|$ **1 mk**

Case 1: $2(10-x) = 5(2x-5) \Rightarrow 20-2x = 10x-25 \Rightarrow x = 15/4$ **1 mk**

Case 2: $2(10-x) = -5(2x-5) \Rightarrow 20-2x = -10x+25 \Rightarrow x = 5/8$ **1 mk**

Solution set = $\{15/4, 5/8\}$ **1 mk**

(b) $5 < 2x+3 < 17 \Rightarrow 1 < x < 7$; **Solution set = $\{x \mid x \in R \wedge 1 < x < 7\}$** **1 mk**

3(iv) — Unit 7: Coordinate Geometry — Rectangle & Isosceles Triangle **5 marks**

Main: Vertices A(0,0), B(6,0), C(6,4), D(0,4)

(a) Compute all four sides:

$AB = \sqrt{(6^2+0^2)} = 6$ $BC = \sqrt{(0^2+4^2)} = 4$ $CD = \sqrt{(6^2+0^2)} = 6$ $DA = \sqrt{(0^2+4^2)} = 4$

Opposite sides equal (AB=CD=6, BC=DA=4) but adjacent sides unequal ⇒ not all sides equal. **2 mk**

(b) Diagonals equal and bisect each other:

$AC = \sqrt{(6^2+4^2)} = \sqrt{52}$ $BD = \sqrt{((-6)^2+4^2)} = \sqrt{52}$ AC = BD √ **1.5 mk**

Midpoint of AC = $((0+6)/2, (0+4)/2) = (3, 2)$

Midpoint of BD = $((6+0)/2, (0+4)/2) = (3, 2)$ Both midpoints equal √

Conclusion: ABCD is a rectangle (equal diagonals that bisect each other). **1.5 mk**

OR — Isosceles Triangle and Median from Apex:

Vertices: P(1,2), Q(7,2), R(4,8)

(a) Side lengths and isosceles check:

$PQ = \sqrt{((7-1)^2+(2-2)^2)} = \sqrt{36} = \mathbf{6}$

$PR = \sqrt{((4-1)^2+(8-2)^2)} = \sqrt{(9+36)} = \sqrt{45} = 3\sqrt{5}$

$QR = \sqrt{((4-7)^2+(8-2)^2)} = \sqrt{(9+36)} = \sqrt{45} = 3\sqrt{5}$

$PR = QR = 3\sqrt{5}$ √ ⇒ **Triangle PQR is isosceles** (two equal sides from apex R). **2.5 mk**

(b) Midpoint M of base PQ; show RM ⊥ PQ:

$M = ((1+7)/2, (2+2)/2) = \mathbf{(4, 2)}$ **1 mk**

Slope of PQ = $(2-2)/(7-1) = 0$ (horizontal)

Slope of RM = $(2-8)/(4-4) = -6/0 \Rightarrow$ undefined (vertical line $x = 4$)

A vertical line is perpendicular to a horizontal line ⇒ **RM ⊥ PQ** √ **1.5 mk**